

Post-hoc Calibration Under Covariate Shift: An Analysis of Classical Tabular Models

Ruililin Shen

bobshenruililil@gmail.com

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Abstract

Temperature scaling and related post-hoc maps are often treated as a cheap fix for miscalibrated probabilities. We measure whether that fix *transfers* when test covariates shift, for logistic regression, random forests, and histogram gradient boosting on small tabular datasets (no deep nets, no API models). On a Gaussian mean-shift of four features at strength 1.5, the difference $\Delta\text{ECE} = \text{ECE}_{\text{shift}} - \text{ECE}_{\text{iid}}$ is positive in 51/60 cells without calibration (mean 0.113, cell-to-cell sd 0.149) and remains positive after i.i.d. temperature, isotonic, or histogram calibration (56, 53, and 57 of 60 cells). A pooled one-sided Wilcoxon on those cells is reported as a *descriptive* summary, not as an independent-sample confirmatory test (results/stats_tests.json contains 20 related p -values with no multiplicity correction). The effect is **not** universal: histogram boosting on breast-cancer has mean $\Delta\text{ECE} = 0.002$ (seed sd 0.009). Post-hoc maps can help or hurt relative to doing nothing (wine random forests: isotonic changes shifted ECE by -0.123; breast-cancer logistic regression: 0.074). Unweighted split-conformal coverage falls with shift strength (breast-cancer logistic regression: 0.913 at $s=0$ to 0.778 at $s=1.5$). We report all seeds, failed hypotheses, and shift-family controls. This is an analysis result, not a new calibrator.

1 Introduction

A classifier is *calibrated* when a predicted probability p is correct with frequency p . Post-hoc maps—Platt-style scaling, histogram binning, isotonic regression, and temperature scaling—are the standard way to improve calibration without retraining [1–3]. A separate literature shows that deep-net uncertainty *collapses* under dataset shift [4–6]. Almost all of that evidence is vision-scale neural nets. Tree ensembles remain strong on tabular data [7], and their i.i.d. calibration properties have been known for decades [2]. Within our API-verified set of 47 papers—mostly deep-net calibration and conformal theory, not a systematic search of applied tabular-drift venues—we did not find this exact measurement (sklearn models, i.i.d. post-hoc maps, explicit $P(X)$ mean-shift). That is a scoped gap, not a proof of absence.

We treat that as a measurement question. We do not in-

troduce a new calibrator. A negative or mixed answer is an acceptable outcome.

Primary hypothesis (H1). Fitting temperature scaling, isotonic regression, or histogram binning on an i.i.d. validation split does not keep ECE from rising when test features are mean-shifted.

Two other hypotheses were piloted and *not* selected as the paper claim: isotonic overfit vs. temperature as a function of n_{cal} (H2; heterogeneous), and oracle-weighted conformal recovery (H3; the 1-D weight estimator failed). Decision log: logs/decisions.md.

2 Related work

Surveys of calibration methods and metrics include [8, 9]. ECE via confidence binning [3, 10] is biased and definition-dependent [11–13]; we therefore also report Brier score [14] and NLL [15]. Temperature scaling [3] and Dirichlet calibration [16] target neural nets. Unsupervised recalibration under covariate shift exists for neural models [17]. Domain-drift post-hoc calibration of CNNs is studied by [6]. Those papers already *predict* our headline outcomes: ECE should rise under shift, and i.i.d.-fitted post-hoc maps should not be a complete fix. A surprising result here would have been that logistic regression, random forests, and histogram boosting *stay* calibrated under a mean-shift of used covariates. We reuse the questions, not the model class.

Conformal prediction has finite-sample coverage under exchangeability [18–20] and requires density-ratio weights under covariate shift [21]. We use unweighted split conformal as a secondary diagnostic, not as a proposed fix.

3 Methods

Models. Scikit-learn logistic regression (with standard scaling), random forests, and HistGradientBoostingClassifier, with hyperparameters as in src/calibshift/models.py. All expose predict_proba.

Calibrators. Fitted on a held-out calibration split of the *source* distribution, then frozen. `none`: identity. `temperature`: one scalar T on $\log p$, fitted by NLL (Guo-style, using log-probabilities so trees without logits still apply). `isotonic`: sklearn isotonic regression on $p(y=1)$ (binary) or on confidence (multiclass). `histogram`: equal-width binning of the same 1-D score.

Shift. Primary: add $s \cdot \hat{\sigma}_j$ to the test-set columns listed in the exp04 config (training and calibration unshifted). Strengths are those in the same config ($s=0$ is i.i.d.). Ablations: quantile slice and exponential-tilt resampling; a control that instead shifts trailing columns (exp07).

Metrics. Equal-width confidence ECE with 15 bins (from the exp04 config), Brier, NLL, accuracy, and split conformal coverage with APS scores $1 - p_y$ at $\alpha = 0.1$.

Protocol. Train/calibration/test partitions follow `calibshift.runner` (source-distribution splits; calibration data never includes the test set), stratified, with the 5 seeds listed in the exp04 config. Results are written only by `calibshift.io.write_result`. `paper/numbers.tex` is generated from those JSON files.

4 Experiments

Datasets. Offline-only: sklearn `breast_cancer` (binary) and `wine` (3-class); two `make_classification` sets (binary and 3-class). No OpenML, no paid APIs.

Main grid (exp04_main_h1). 240 cells, 0 failed, 23.0 seconds on CPU. This is the source of all headline numbers.

Ablations. exp05: quantile slice and importance resampling. exp06: calibration-set size (killed primary H2). exp07: shift trailing columns.

Sanity. I.i.d. logistic regression on breast-cancer: accuracy 0.979, ECE 0.032 ± 0.010 (linear models on this set are typically $\gtrsim 0.95$ accurate). Wine random forests are overconfident i.i.d. (ECE 0.147 ± 0.020), matching the classical tree-calibration picture [2]. Split conformal coverage at $s=0$ is near the nominal $1 - \alpha = 0.9$ for $\alpha=0.1$. We do *not* compare ECE magnitudes to ImageNet temperature scaling [3].

5 Results

5.1 H1: ECE usually rises under mean-shift

Figure 1 plots shifted ECE against s . Figure 2 shows mean ΔECE at $s=1.5$. Pooled over 60 dataset $\times$ model $\times$ seed cells, uncalibrated mean $\Delta\text{ECE} = 0.113$ with cell-to-cell sd 0.149 (51 positive, 9 negative). After i.i.d. temperature scaling the pooled mean is 0.124 (sd 0.150). Isotonic and histogram are

Table 1: Pooled ΔECE at $s=1.5$ ($n = 60$ cells: 4 datasets $\times$ 3 models $\times$ 5 seeds). “std” is the sample sd of the 60 cell differences (heterogeneity), not se of the mean. Wilcoxon p is one-sided and *unadjusted* among 20 related tests in `results/stats_tests.json`.

Calibrator	mean ΔECE	std	#pos/neg	Wilcoxon p
none	0.113	0.149	51/9	2.2e-10
temperature	0.124	0.150	56/4	3.1e-11
isotonic	0.141	0.151	53/7	4.6e-11
histogram	0.141	0.151	57/3	3.3e-11

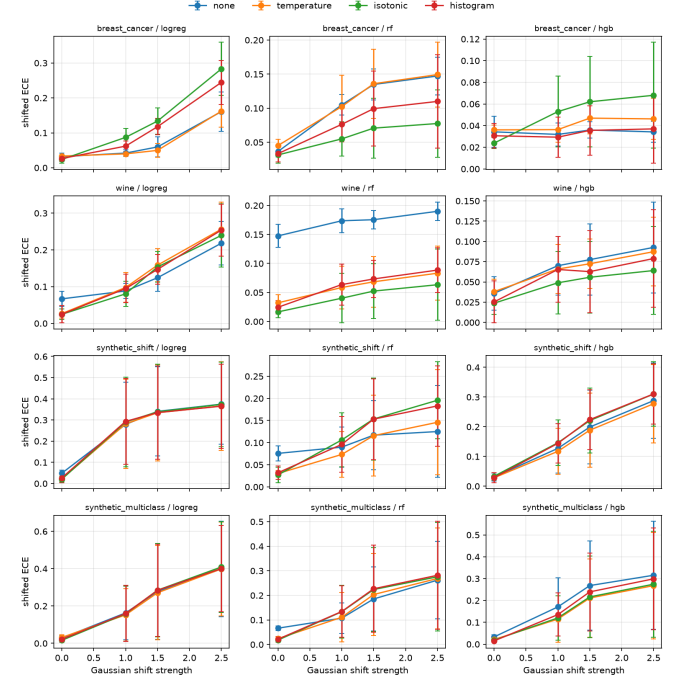


Figure 1: Shifted ECE vs. Gaussian mean-shift strength (mean $\pm$ sd over 5 seeds). $s=0$ is i.i.d. Source: `results/exp04_main_h1.json`.

similar (0.141 and 0.141). These sd figures describe heterogeneity across cells, not a confidence interval for the mean. A one-sided Wilcoxon signed-rank statistic on the pooled cells is exploratory: the 60 differences are not independent replicates (models share seeds and datasets). Synthetic logistic regression is the largest effect (mean 0.292, sd 0.214); breast-cancer histogram boosting is the exception (mean 0.002, seed sd 0.009; this is a descriptive spread, not an interval that “includes 0”). Table 1 records the pooled statistics.

5.2 Post-hoc maps are not a reliable remedy

Figure 3 plots $\text{ECE}_{\text{shift}}(\text{cal}) - \text{ECE}_{\text{shift}}(\text{none})$. Wine random forests *benefit* from isotonic (-0.123 ECE; shifted ECE $0.175 \rightarrow 0.068$), because the uncalibrated forest was already badly calibrated. Breast-cancer logistic regression is *hurt* by isotonic (0.074). We therefore do not rank calibrators under shift.

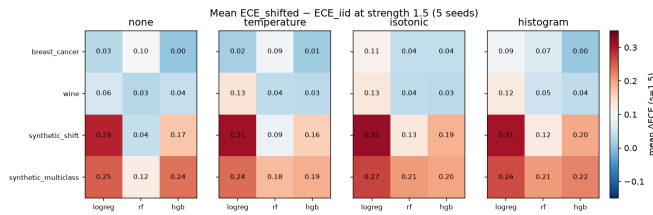


Figure 2: Mean ΔECE at $s=1.5$. Positive (red) means the shifted test is *worse* calibrated than the i.i.d. test.

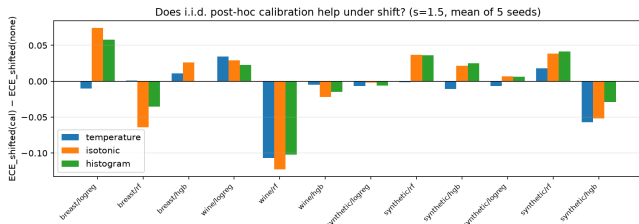


Figure 3: Does i.i.d. calibration help on the shifted test? Negative bars mean the calibrator beats *none* under shift.

5.3 Coverage tracks broken exchangeability

Unweighted split conformal (logreg, *none*) coverage at $\alpha = 0.1$: breast-cancer 0.913 ($s=0$), 0.778 ($s=1.5$), 0.608 ($s=2.5$); synthetic binary 0.634 at $s=1.5$ (Figure 4). This is predicted by theory [21]; we are not claiming a new conformal method. A 1-D “oracle” density ratio on feature 0 did *not* restore coverage in the H3 pilot (killed).

5.4 Controls

Quantile slicing induces small ΔECE relative to the Gaussian mean-shift (exp05 JSON). Importance resampling can *decrease* ECE on breast-cancer (the resampled test is not harder). Shifting trailing columns on *synthetic_shift* yields RF $\Delta\text{ECE} = 0.009$ vs. 0.041 when shifting the first four columns: trees are less sensitive to unused coordinates; logistic regression is not. H2 (isotonic vs. temperature vs. n_{cal}) remains mixed and is not a paper claim.

6 Limitations

- **Shift is synthetic.** A four-feature mean-shift is not WILDS [22] and is entangled with accuracy drop (synthetic logreg accuracy 0.932 $\rightarrow$ 0.620 at $s=1.5$).
- **Small tabular data only.** No ImageNet, no language models.
- **Multiclass isotonic/histogram map confidence,** not a full Dirichlet map [16].
- **ECE estimator.** Equal-width bins (15, from the exp04 config); we also logged quantile-binned ECE in the JSON but did not promote it to a claim.

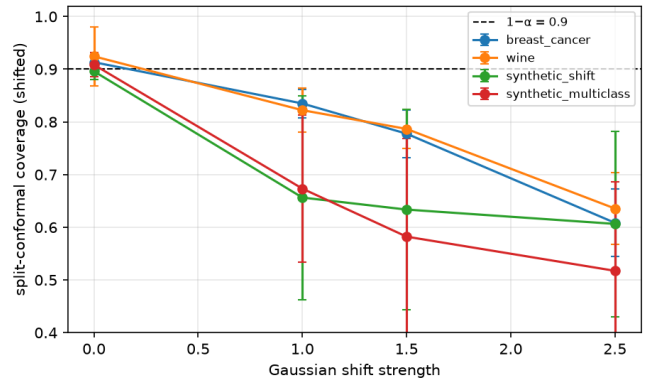


Figure 4: Unweighted split-conformal coverage vs. shift (logreg, no calibrator). Dashed line: $1 - \alpha = 0.9$.

- **Temperature on probabilities.** Trees have no logits; we scale $\log p$. That is not identical to logit-temperature on a neural net.
- **H3 weights were misspecified.** A 1-D histogram ratio is not $p_{\text{test}}(x)/p_{\text{cal}}(x)$ for a 4-D shift.
- **Pooling and tests.** The headline Wilcoxon pools heterogeneous dataset $\times$ model cells that are not independent. We ran 20 one-sided tests with no multiplicity correction; p -values are exploratory. Per-dataset tests are in `results/stats_tests.json`.
- **Search scope.** Novelty is relative to `paper/verified.bib` (47 API-verified papers), not a systematic review of applied tabular-drift venues.

7 Conclusion

For sklearn tabular models, i.i.d. post-hoc calibration is not a dependable answer to covariate shift of the kind we applied. ECE usually rises, conformal coverage falls, and the popular maps can make shifted ECE better *or* worse than doing nothing. The honest exception (breast-cancer HGB) and the failed H2/H3 hypotheses are part of the result.

AI-generation disclosure

This repository and draft were produced by a coding agent executing the user’s GOAL.md protocol (literature verification via arXiv and Semantic Scholar APIs, experiments on CPU, no hand-edited `results/*.json`). Every numeric macro in `paper/numbers.tex` is generated from those files. Citations are restricted to `paper/verified.bib`. A human should still read the limitations and re-run `make fresh-clone-test` before treating the PDF as camera-ready.

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